

Q1. List top 5 customers by total order amount.

**Retrieve the top 5 customers who have spent the most across all sales orders.
Show CustomerID, CustomerName, and TotalSpent.**

Query

```
SELECT TOP 5
    c.CustomerID,
    c.Name AS CustomerName,
    SUM(so.TotalAmount) AS TotalSpent
FROM dbo.SalesOrder so
JOIN dbo.Customer c ON so.CustomerID = c.CustomerID
GROUP BY c.CustomerID, c.Name
ORDER BY TotalSpent DESC;
```

Output

	CustomerID	CustomerName	TotalSpent
1	98	Amando	232783.99
2	12	Dwight	175719.20
3	84	Courtney	156259.77
4	90	Chelsae	149290.57
5	27	Lawton	141343.95

Q2. Find the number of products supplied by each supplier.

Display SupplierID, SupplierName, and ProductCount. Only include suppliers that have more than 10 products.

Query

```
SELECT
    s.SupplierID,
    s.Name AS SupplierName,
    COUNT(DISTINCT pod.ProductID) AS ProductCount
FROM dbo.Supplier s
JOIN dbo.PurchaseOrder po
    ON s.SupplierID = po.SupplierID
JOIN dbo.PurchaseOrderDetail pod
    ON po.OrderID = pod.OrderID
GROUP BY s.SupplierID, s.Name
HAVING COUNT(DISTINCT pod.ProductID) > 10;
```

Output

	SupplierID	SupplierName	ProductCount
1	1	Schamberger Inc	28
2	2	Bogisch-Block	30
3	3	Abshire Inc	29
4	4	Schmidt LLC	29
5	5	Mann, Will and...	24
6	6	Abemathy-Wa...	26
7	7	Kassulke LLC	32
8	8	Lang-Rodriguez	28
9	9	Sawayn-Adams	28
10	10	Stokes, Funk a...	28

Q3. Identify products that have been ordered but never returned.
Show ProductID, ProductName, and total order quantity.

Query

```
SELECT
    p.ProductID,
    p.Name AS ProductName,
    SUM(sod.Quantity) AS TotalOrderQuantity
FROM dbo.Product p
JOIN dbo.SalesOrderDetail sod
    ON p.ProductID = sod.ProductID
WHERE NOT EXISTS (
    SELECT 1
    FROM dbo.ReturnDetail rd
    WHERE rd.ProductID = p.ProductID
)
GROUP BY p.ProductID, p.Name;
```

Output

	ProductID	ProductName	TotalOrderQuantity
1	3	Sertraline Hydrochloride	20
2	15	Flecainide Acetate	55

Q4. For each category, find the most expensive product.
Display CategoryID, CategoryName, ProductName, and Price. Use a subquery to get the max price per category.

Query

```
SELECT
    c.CategoryID,
    c.Name AS CategoryName,
    p.Name AS ProductName,
    p.Price
FROM dbo.Product p
JOIN dbo.Category c
    ON p.CategoryID = c.CategoryID
JOIN (
    SELECT
        CategoryID,
        MAX(Price) AS MaxPrice
    FROM dbo.Product
    GROUP BY CategoryID
) max_p
    ON p.CategoryID = max_p.CategoryID
   AND p.Price = max_p.MaxPrice;
```

Output

	CategoryID	CategoryName	ProductName	Price
1	20	Hard Tile & Stone	Butalbital, Acetaminophen and Caffeine	85.95
2	19	Epoxy Flooring	Entex T	201...
3	18	Wall Protection	Natural Cherry Honey Herb Throat Dr...	250...
4	17	Termite Control	SohMed Cold Relief	281...
5	16	Elevator	Sertraline Hydrochloride	289...
6	15	Hard Tile & Stone	Quetiapine Fumarate	183...
7	14	Glass & Glazing	Losartan Potassium	274...
8	13	NULL	COMETRIQ	277...
9	12	Asphalt Paving	Acetaminophen PM	267...
10	10	Exterior Signage	Flecainide Acetate	232...
11	9	Termite Control	Secura Moisturizing Cleanser	262...
12	7	Fire Protection	Oral Analgesic	298...
13	6	Electrical and Fi...	Fluoxetine	726...
14	5	Soft Flooring an...	Isopropyl Alcohol Surgical Scrub	255...
15	4	Painting & Vinyl ...	Focalin	245...
16	3	Framing (Wood)	ILLUMINATING DAILY MOISTURIZ...	243...
17	2	Landscaping & ...	Betapace AF	154...

Q5. List all sales orders with customer name, product name, category, and supplier.

For each sales order, display:

OrderID, CustomerName, ProductName, CategoryName, SupplierName, and Quantity.

Query

```
SELECT
    so.OrderID,
    c.Name AS CustomerName,
    p.Name AS ProductName,
    cat.Name AS CategoryName,
    s.Name AS SupplierName,
    sod.Quantity
FROM dbo.SalesOrder so
JOIN dbo.Customer c
    ON so.CustomerID = c.CustomerID
JOIN dbo.SalesOrderDetail sod
    ON so.OrderID = sod.OrderID
JOIN dbo.Product p
    ON sod.ProductID = p.ProductID
JOIN dbo.Category cat
    ON p.CategoryID = cat.CategoryID
JOIN dbo.PurchaseOrderDetail pod
    ON p.ProductID = pod.ProductID
JOIN dbo.PurchaseOrder po
    ON pod.OrderID = po.OrderID
JOIN dbo.Supplier s
    ON po.SupplierID = s.SupplierID;
```

Output

	OrderID	CustomerName	ProductName	CategoryName	SupplierName	Quantity
1	21	Marcela	Fluoxetine	Electrical and Fire Alarm	Schmidt LLC	31
2	27	Sayres	Fluoxetine	Electrical and Fire Alarm	Schmidt LLC	12
3	33	Kristina	Fluoxetine	Electrical and Fire Alarm	Schmidt LLC	11
4	9	Evvie	Dexamethasone	Landscaping & Irrigation	Kassulke LLC	25
5	76	Kristina	Sertraline Hydrochloride	Elevator	Bogisch-Block	20
6	29	Oriana	SALSOLA KALI POLLEN	Fire Protection	Abshire Inc	37
7	20	Oriana	COMETRIQ	NULL	Mann, Will and Fritsch	15
8	29	Oriana	SALSOLA KALI POLLEN	Fire Protection	Bogisch-Block	37
9	18	Lawton	Isopropyl Alcohol Surgical Scrub	Soft Flooring and Base	Mann, Will and Fritsch	32
10	36	Lucais	Isopropyl Alcohol Surgical Scrub	Soft Flooring and Base	Mann, Will and Fritsch	28
11	62	Eddy	Isopropyl Alcohol Surgical Scrub	Soft Flooring and Base	Mann, Will and Fritsch	17
12	73	Courtney	Isopropyl Alcohol Surgical Scrub	Soft Flooring and Base	Mann, Will and Fritsch	40
13	87	Lynn	Isopropyl Alcohol Surgical Scrub	Soft Flooring and Base	Mann, Will and Fritsch	40
14	44	Xenia	SohMed Cold Relief	Termite Control	Schmidt LLC	14
15	55	Lawrence	SohMed Cold Relief	Termite Control	Schmidt LLC	11
16	70	Pattie	SohMed Cold Relief	Termite Control	Schmidt LLC	34
17	72	Huntington	SohMed Cold Relief	Termite Control	Schmidt LLC	24
18	14	Neils	Acetaminophen PM	Asphalt Paving	Bogisch-Block	31
19	41	Lief	Acetaminophen PM	Asphalt Paving	Bogisch-Block	20
20	98	Andonis	Acetaminophen PM	Asphalt Paving	Bogisch-Block	39
21	57	Lucais	Icy Hot Back and Large Areas	Asphalt Paving	Abemathy-Waelchi	11
22	59	Lucas	Icy Hot Back and Large Areas	Asphalt Paving	Abemathy-Waelchi	15

Q6. Find all shipments with details of warehouse, manager, and products shipped.

Display:

ShipmentID, WarehouseName, ManagerName, ProductName, QuantityShipped, and TrackingNumber.

Query

```

SELECT
    s.ShipmentID,
    l.Name AS WarehouseName,
    e.Name AS ManagerName,
    p.Name AS ProductName,
    sd.Quantity AS QuantityShipped,
    s.TrackingNumber
FROM dbo.Shipment s
JOIN dbo.ShipmentDetail sd
    ON s.ShipmentID = sd.ShipmentID
JOIN dbo.Product p
    ON sd.ProductID = p.ProductID
JOIN dbo.Warehouse w
    ON s.WarehouseID = w.WarehouseID
JOIN dbo.Location l
    ON w.LocationID = l.LocationID
JOIN dbo.Employee e
    ON w.ManagerID = e.EmployeeID;

```

Output

	ShipmentID	WarehouseName	ManagerName	ProductName	QuantityShipped	TrackingNumber
1	1	Ms	Daloris	Midol Complete	23	NULL
2	2	Rev	Rose	Quetiapine F...	25	NULL
3	3	Ms	Daloris	Secura Moist...	23	NULL
4	4	Rev	Rose	Entex T	14	NULL
5	5	Ms	Daloris	Promethazine...	29	NULL
6	6	Mr	Neille	PASPALUM ...	6	NULL
7	7	Mr	Aurora	SHISEIDO A...	1	NULL
8	8	Mr	Neille	Icy Hot Back...	8	NULL
9	9	Rev	Rose	SHISEIDO A...	7	NULL
10	10	Rev	Rose	COMETRIQ	14	NULL
11	11	Mr	Aurora	Natural Cherr...	20	NULL
12	12	Rev	Rose	Icy Hot Back...	30	NULL
13	13	Ms	Daloris	ILLUMINATI...	18	NULL
14	14	Ms	Daloris	Midol Complete	18	NULL
15	15	Mr	Aurora	Maximum Str...	15	NULL
16	16	Ms	Daloris	Oral Analgesic	17	NULL
17	17	Mr	Rose	Entex T	22	NULL
18	18	Mr	Neille	Quetiapine F...	21	NULL
19	19	Ms	Daloris	Dexamethas...	14	NULL
20	20	Rev	Rose	Sertraline Hy...	6	NULL
21	21	Mrs	Basilio	Antibacterial ...	21	NULL
22	22	Mr	Neille	PASPALUM ...	1	NULL
23	23	Ms	Daloris	SohMed Cold...	22	NULL
24	24	Mr	Aurora	Granisetron ...	30	NULL

Q7. Find the top 3 highest-value orders per customer using RANK(). Display CustomerID, CustomerName, OrderID, and TotalAmount.

Query

```
WITH RankedOrders AS (
    SELECT
        c.CustomerID,
        c.Name AS CustomerName,
        so.OrderID,
        so.TotalAmount,
        RANK() OVER (
            PARTITION BY c.CustomerID
            ORDER BY so.TotalAmount DESC
        ) AS OrderRank
    FROM dbo.Customer c
    JOIN dbo.SalesOrder so
    ON c.CustomerID = so.CustomerID
)
SELECT *
FROM RankedOrders
WHERE OrderRank <= 3;
```

Output

	CustomerID	CustomerName	OrderID	TotalAmount	OrderRank
1	2	Daune	97	29549.27	1
2	3	Nevile	68	53370.84	1
3	3	Nevile	93	46081.93	2
4	4	Eustace	3	29705.50	1
5	6	Murray	16	12849.15	1
6	7	Lem	91	73192.12	1
7	11	Dory	43	51043.81	1
8	12	Dwight	5	91621.22	1
9	12	Dwight	52	59504.03	2
10	12	Dwight	6	24593.95	3
11	13	Liliane	46	44724.06	1
12	16	Sayres	27	22896.23	1
13	19	Xenia	44	76595.26	1
14	19	Xenia	74	61123.59	2
15	21	Lucais	36	50207.49	1
16	21	Lucais	57	44102.74	2
17	24	Eddy	62	80551.84	1
18	25	Lief	41	58133.77	1
19	25	Lief	47	32650.50	2
20	26	Talbot	66	62369.56	1

Q8. For each product, show its sales history with the previous and next sales quantities (based on order date). Display ProductID, ProductName, OrderID, OrderDate, Quantity, PrevQuantity, and NextQuantity.

Query

```

SELECT
    p.ProductID,
    p.Name AS ProductName,
    so.OrderID,
    so.OrderDate,
    sod.Quantity,
    LAG(sod.Quantity) OVER (PARTITION BY p.ProductID ORDER BY so.OrderDate, so.OrderID) AS PrevQuantity,
    LEAD(sod.Quantity) OVER (PARTITION BY p.ProductID ORDER BY so.OrderDate, so.OrderID) AS NextQuantity
FROM
    dbo.Product p
INNER JOIN
    dbo.SalesOrderDetail sod ON p.ProductID = sod.ProductID
INNER JOIN
    dbo.SalesOrder so ON sod.OrderID = so.OrderID;

```

Output

	ProductID	ProductName	OrderID	OrderDate	Quantity	PrevQuantity	NextQuantity
1	1	Butalbital, Acetaminophen and Caffeine	79	2003-04-21 04:11:35	19	NULL	17
2	1	Butalbital, Acetaminophen and Caffeine	61	2016-05-18 03:37:40	17	19	13
3	1	Butalbital, Acetaminophen and Caffeine	88	2017-07-11 14:02:27	13	17	NULL
4	2	SHISEIDO ADVANCED HYDRO-LIQ...	30	2000-12-26 15:50:29	23	NULL	NULL
5	3	Sertraline Hydrochloride	76	2009-10-24 02:52:28	20	NULL	NULL
6	4	SohMed Cold Relief	72	2003-11-14 11:53:28	24	NULL	14
7	4	SohMed Cold Relief	44	2004-04-04 09:26:01	14	24	11
8	4	SohMed Cold Relief	55	2012-04-24 07:46:10	11	14	34
9	4	SohMed Cold Relief	70	2021-08-30 09:18:44	34	11	NULL
10	5	Promethazine Hydrochloride	15	2000-05-29 09:37:55	13	NULL	23
11	5	Promethazine Hydrochloride	80	2017-08-05 16:48:19	23	13	NULL
12	6	Ondansetron	23	2002-02-06 08:48:16	14	NULL	24
13	6	Ondansetron	86	2011-05-18 11:07:42	24	14	37
14	6	Ondansetron	53	2013-01-29 02:06:34	37	24	35
15	6	Ondansetron	50	2020-10-08 05:33:26	35	37	NULL
16	7	TopCare Antacid Calcium	43	2007-12-19 07:08:49	15	NULL	29
17	7	TopCare Antacid Calcium	94	2008-03-01 03:58:21	29	15	NULL
18	8	Maximum Strength	17	2015-07-29 20:13:43	21	NULL	NULL

Q9. Create a view named vw_CustomerOrderSummary that shows for each customer:

CustomerID, CustomerName, TotalOrders, TotalAmountSpent, and LastOrderDate.

Query

```
CREATE OR ALTER VIEW dbo.vw_CustomerOrderSummary AS
SELECT
    c.CustomerID,
    c.Name AS CustomerName,
    COUNT(so.OrderID) AS TotalOrders,
    ISNULL(SUM(so.TotalAmount), 0) AS TotalAmountSpent,
    MAX(so.OrderDate) AS LastOrderDate
FROM dbo.Customer c
LEFT JOIN dbo.SalesOrder so
    ON c.CustomerID = so.CustomerID
GROUP BY c.CustomerID, c.Name;

SELECT * FROM dbo.vw_CustomerOrderSummary;
```

Output

Results		Messages			
	CustomerID	CustomerName	TotalOrders	TotalAmountSpent	LastOrderDate
1	1	Pavla	0	0.00	NULL
2	2	Daune	1	29549.27	2008-12-11 11:36:26
3	3	Nevile	2	99452.77	2010-11-30 13:33:08
4	4	Eustace	1	29705.50	2019-12-22 19:57:18
5	5	Elga	0	0.00	NULL
6	6	Murray	1	12849.15	2008-07-06 01:10:56
7	7	Lem	1	73192.12	2012-03-18 16:38:10
8	8	Reube	0	0.00	NULL
9	9	Sergeant	0	0.00	NULL
10	10	Valeria	0	0.00	NULL
11	11	Dory	1	51043.81	2007-12-19 07:08:49
12	12	Dwight	3	175719.20	2020-07-21 05:01:11
13	13	Liliane	1	44724.06	2012-12-21 02:35:15
14	14	Juditha	0	0.00	NULL
15	15	Claudianus	0	0.00	NULL
16	16	Sayres	1	22896.23	2022-11-16 00:14:29
17	17	Leslie	0	0.00	NULL
18	18	Waring	0	0.00	NULL
19	19	Xenia	2	137718.85	2007-06-18 06:30:46
20	20	Jemaine	0	0.00	NULL

Q10. Write a stored procedure `sp_GetSupplierSales` that takes a `SupplierID` as input and returns the total sales amount for all products supplied by that supplier.

Query

```
CREATE OR ALTER PROCEDURE dbo.sp_GetSupplierSales
    @SupplierID INT
AS
BEGIN
    SET NOCOUNT ON;

    SELECT
        s.SupplierID,
        s.Name AS SupplierName,
        ISNULL(SUM(sod.TotalAmount), 0) AS TotalSalesAmount
    FROM dbo.Supplier s
    JOIN dbo.PurchaseOrder po
        ON s.SupplierID = po.SupplierID
    JOIN dbo.PurchaseOrderDetail pod
        ON po.OrderID = pod.OrderID
    JOIN dbo.SalesOrderDetail sod
        ON pod.ProductID = sod.ProductID
    JOIN dbo.SalesOrder so
        ON sod.OrderID = so.OrderID
    WHERE s.SupplierID = @SupplierID
    GROUP BY s.SupplierID, s.Name;
END;
EXEC dbo.sp_GetSupplierSales @SupplierID = 2;
```

Output

Results		Messages	
	SupplierID	SupplierName	TotalSalesAmount
1	2	Bogisich-Block	294969.99